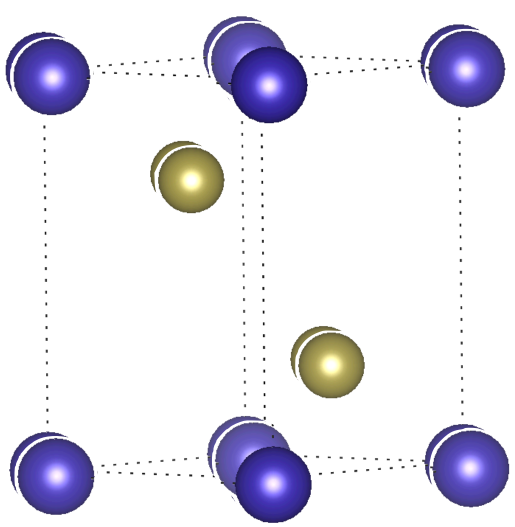
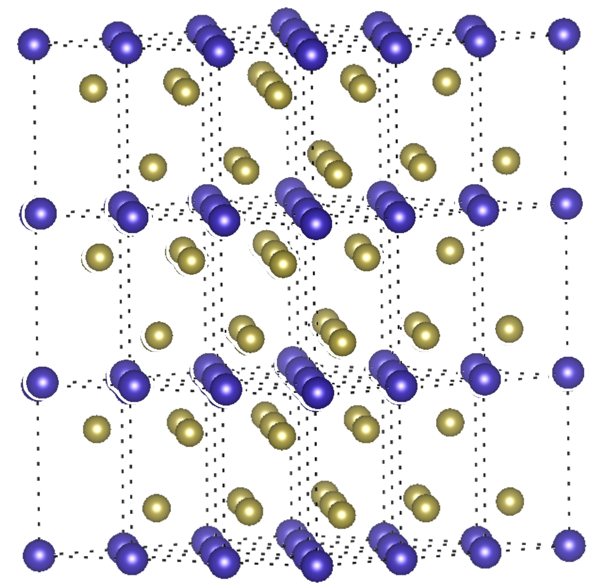
DFT: 2D Materials

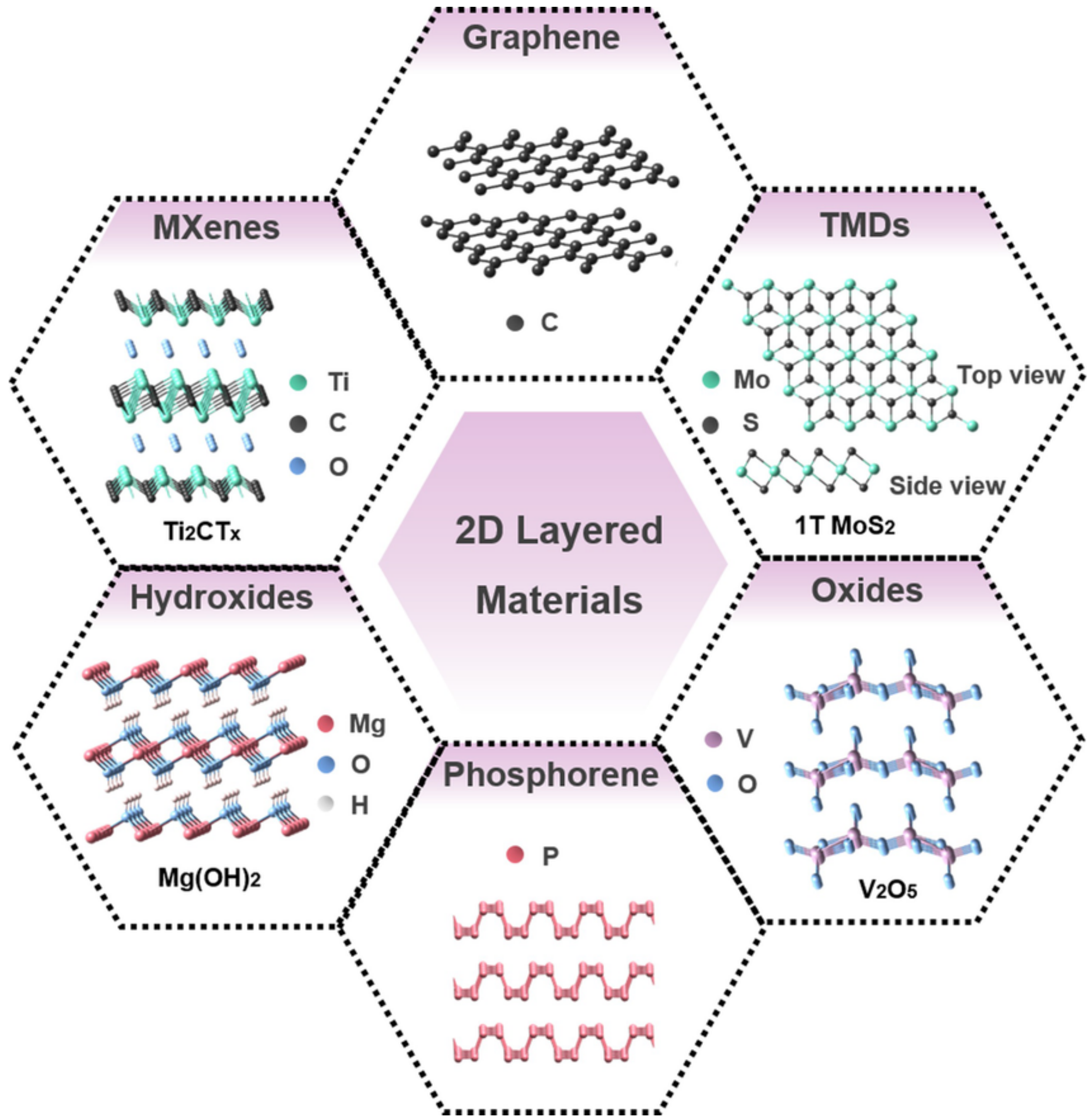
From Bulks to Layers.

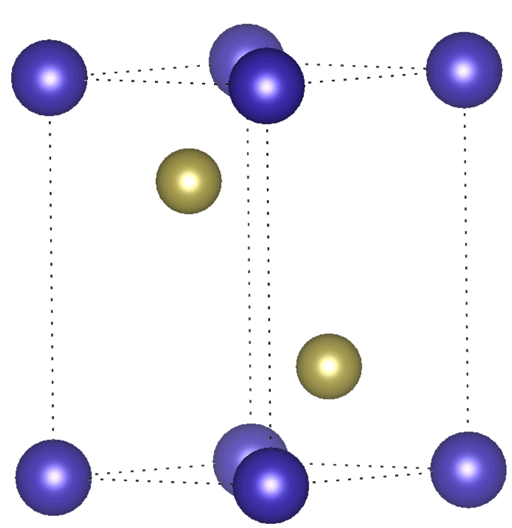
Nicolás F. Barrera - Javiera Cabezas-Escares - Andrea Echeverri

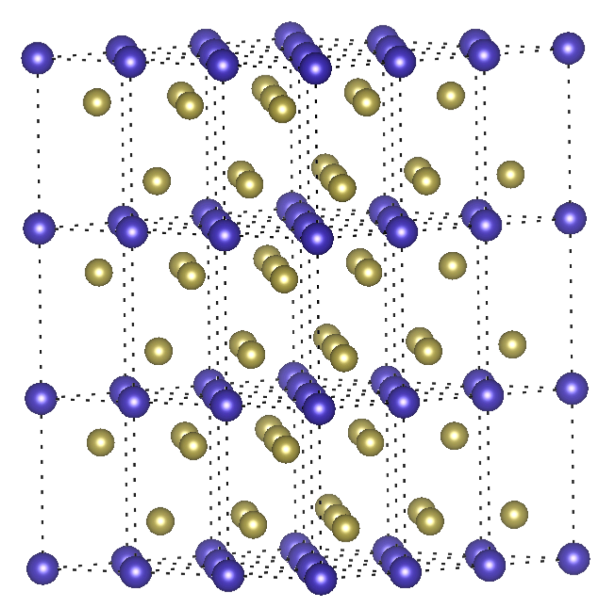
From Bulk to Slides

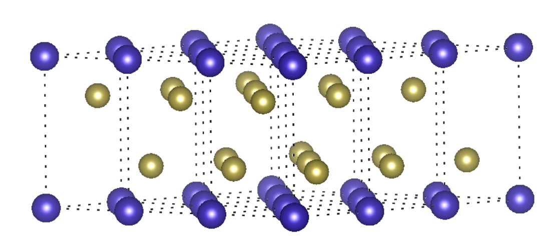




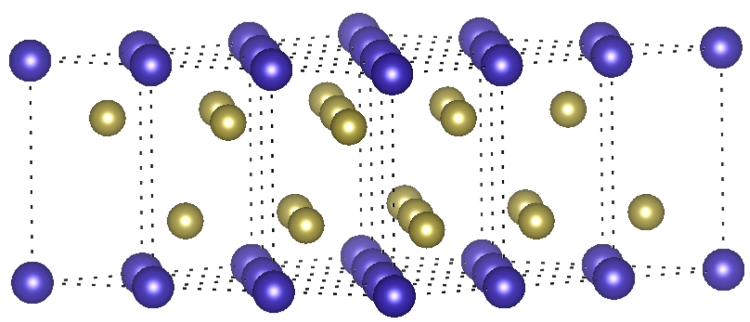

Graphene
C
MXenes
Ti
C
O
Ti2CTx
TMDs
Mo
S
Top view
Side view
1T MoS2
2D Layered
Materials
Hydroxides
Mg
O
H
Mg(OH)2
Oxides
V
O
V2O5
Phosphorene
P

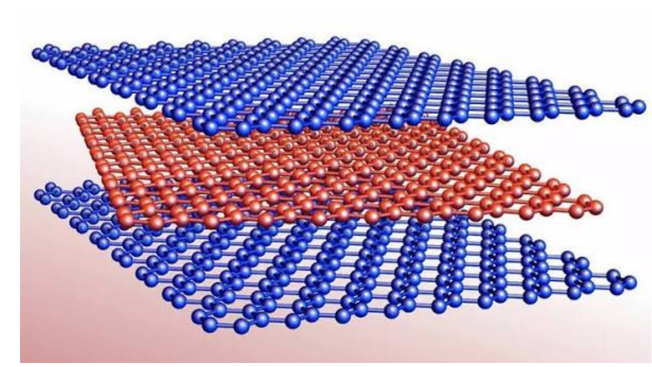


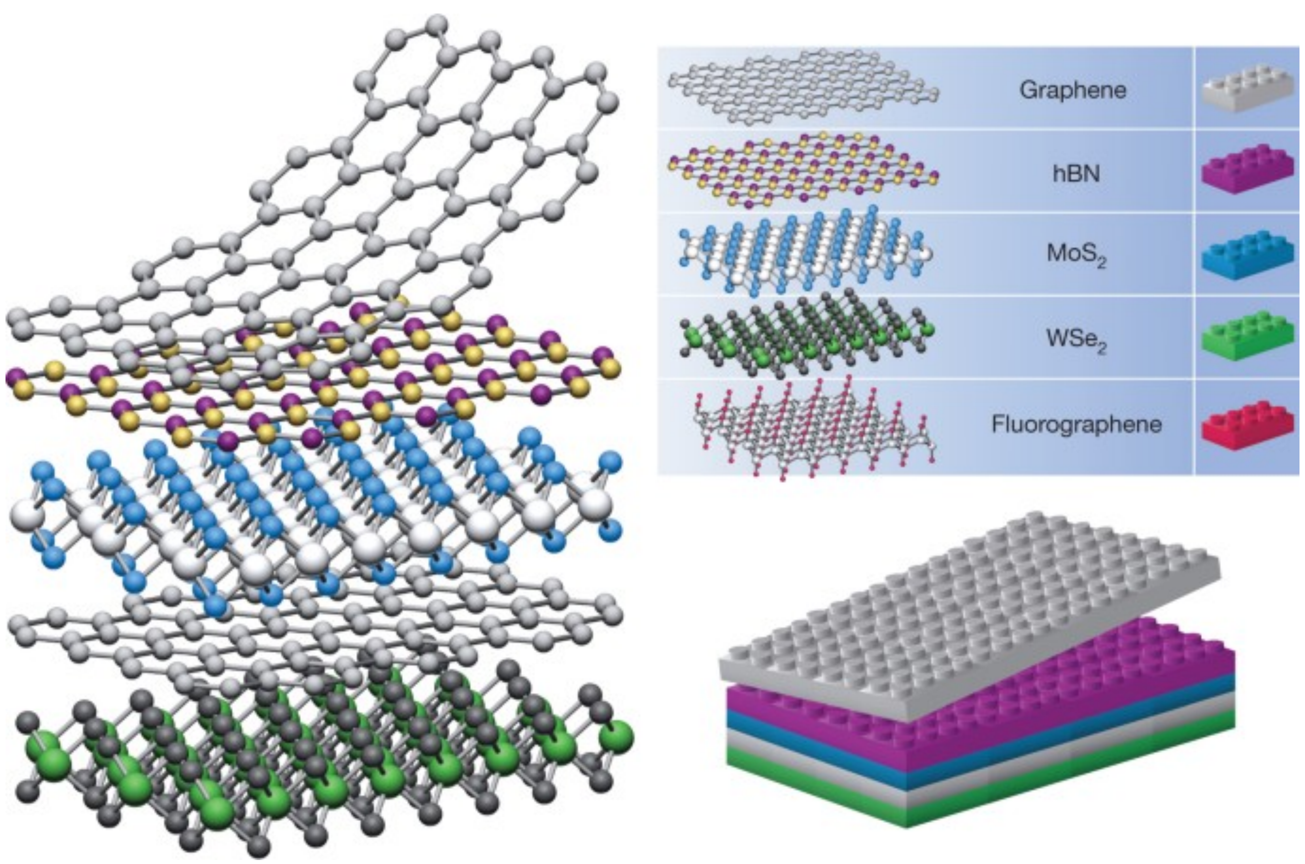


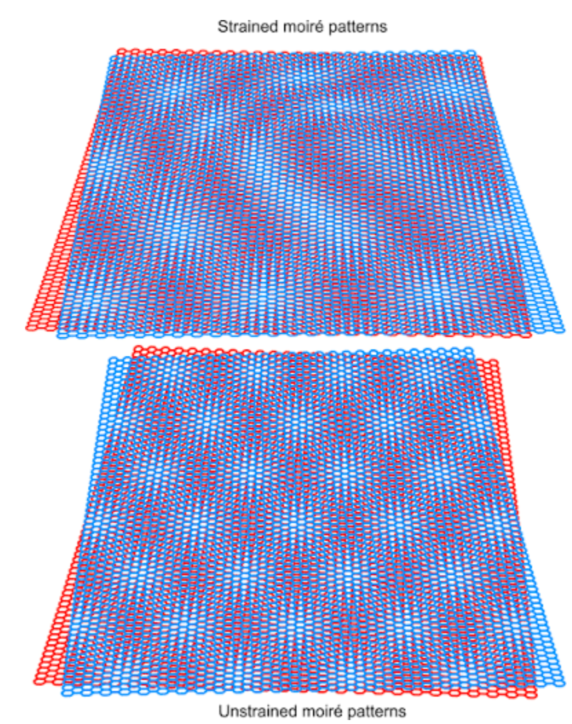


New Frontiers: vdW Structures and Twistronic

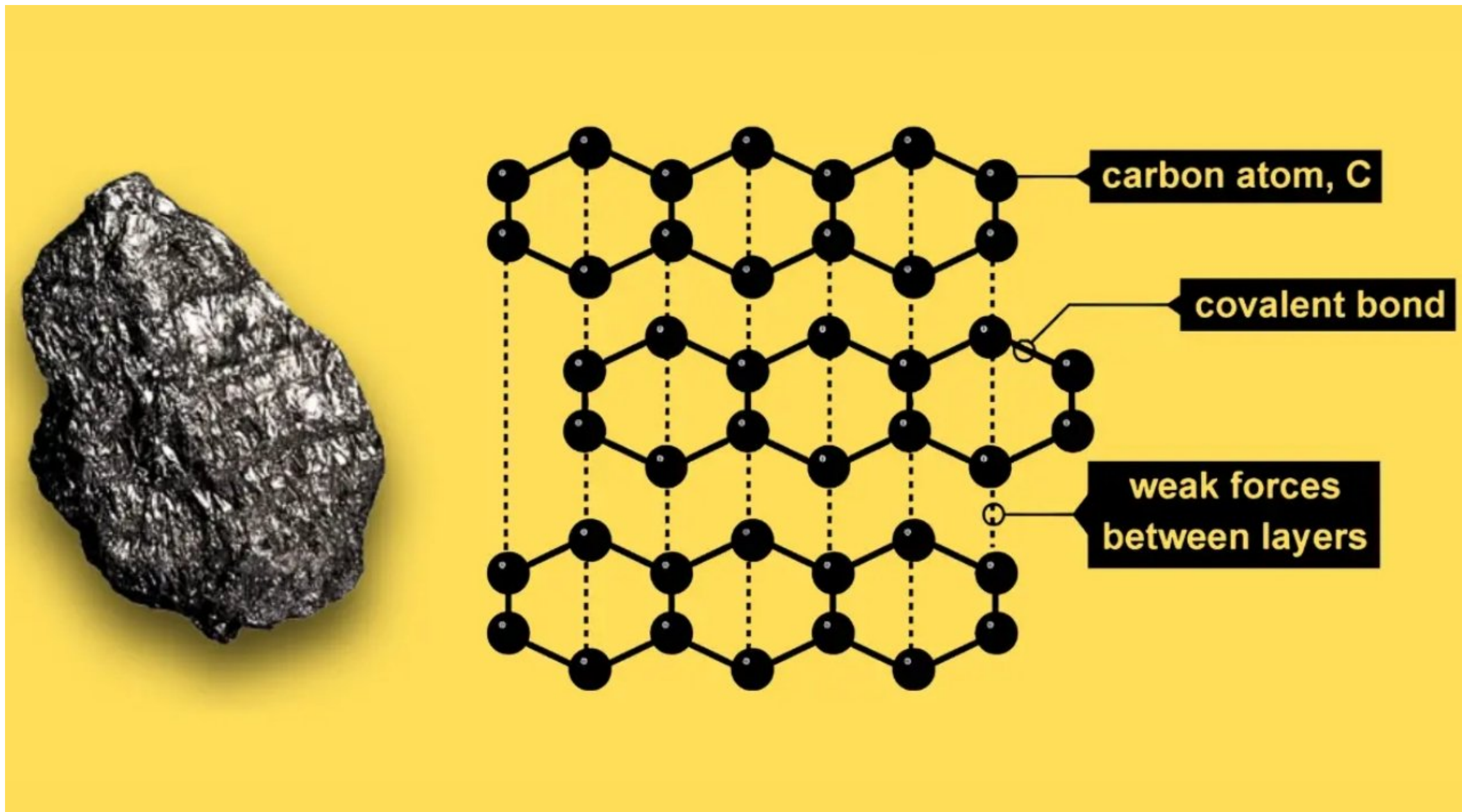


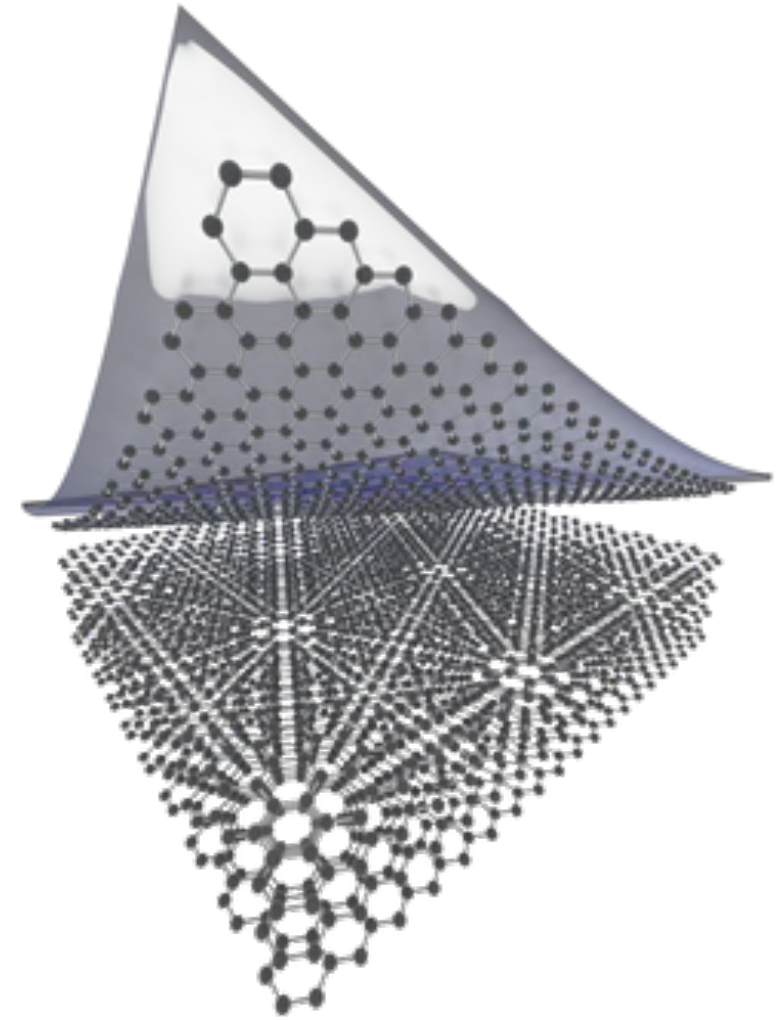


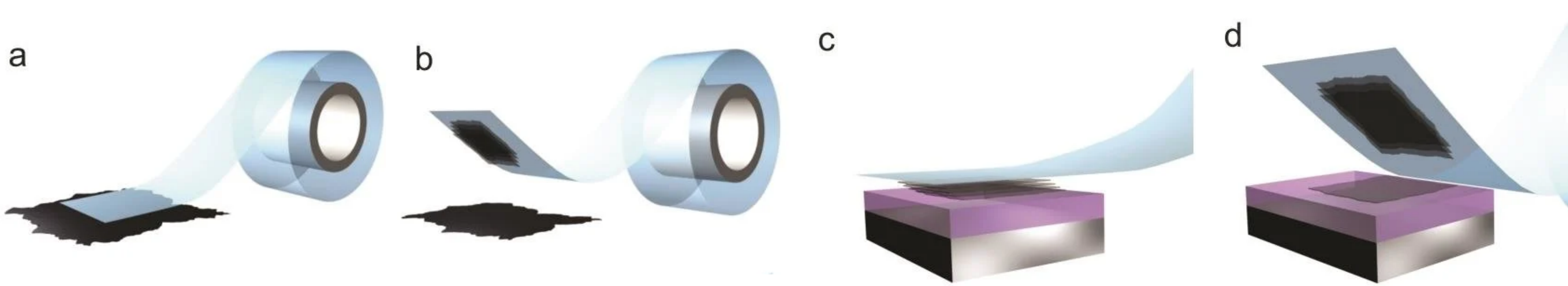




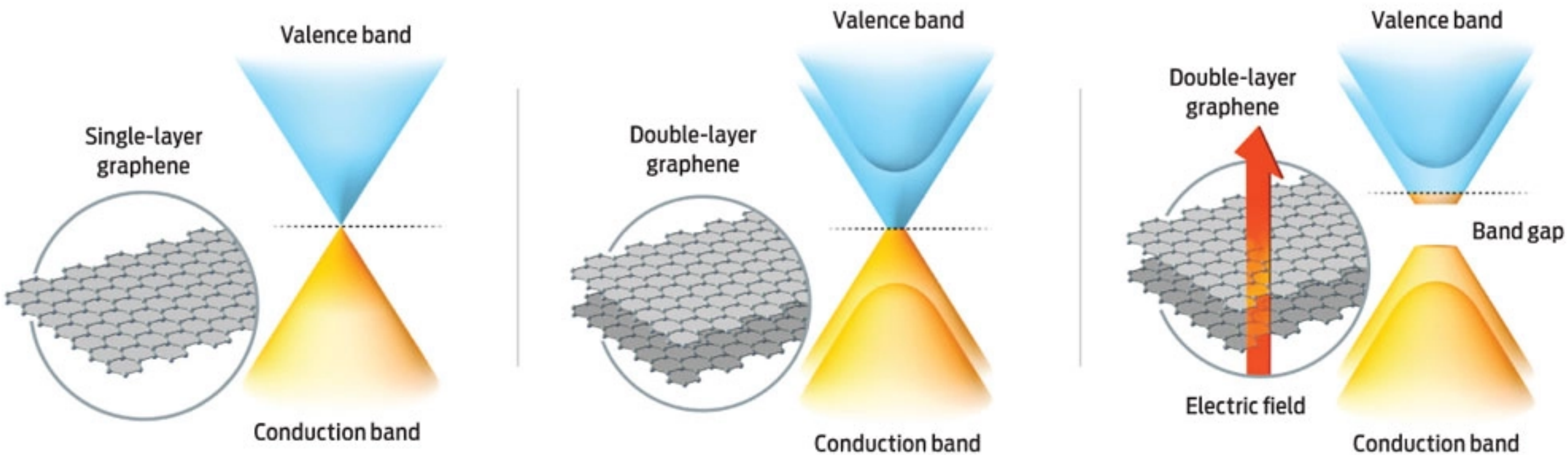
From Graphite to Graphene

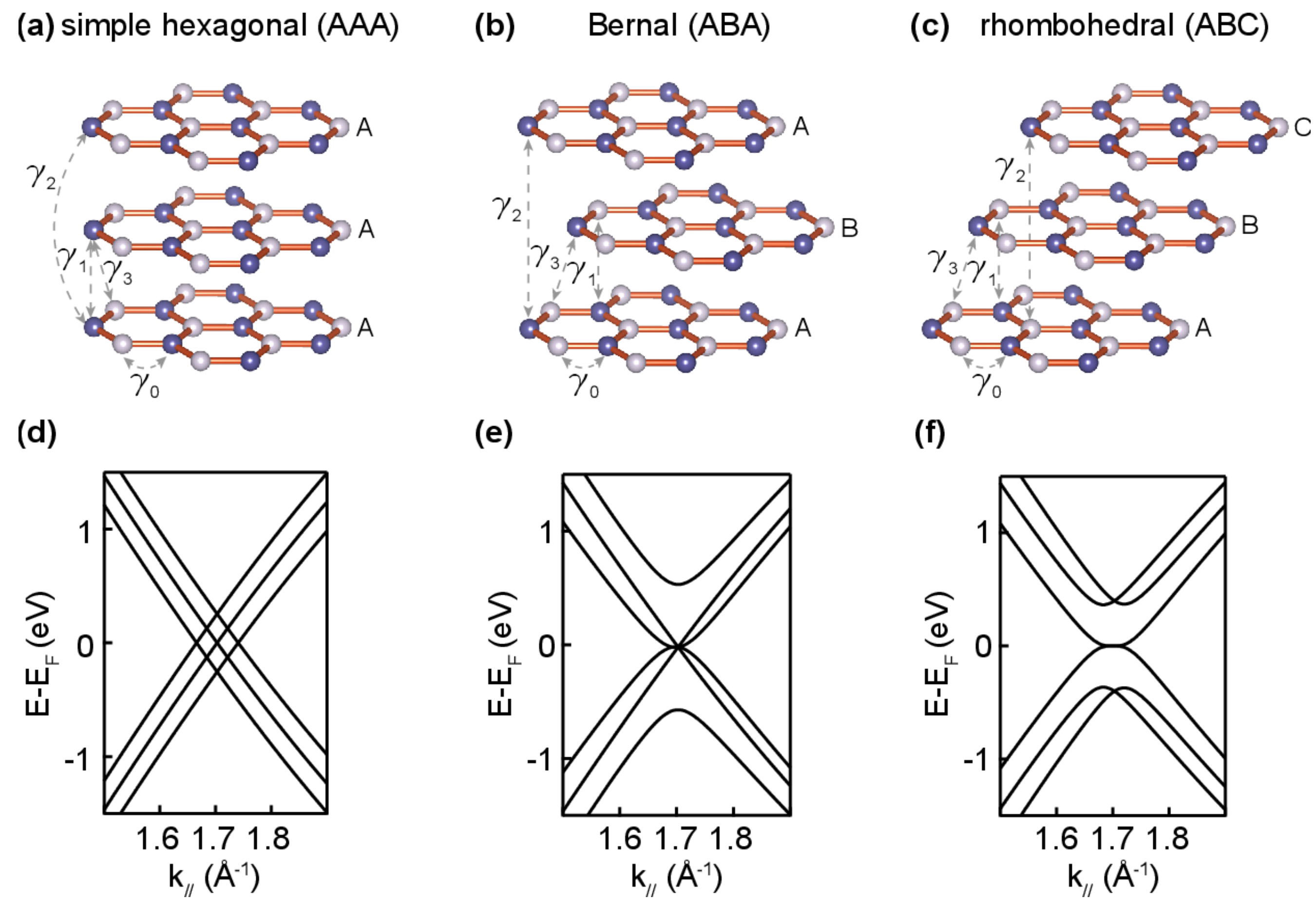


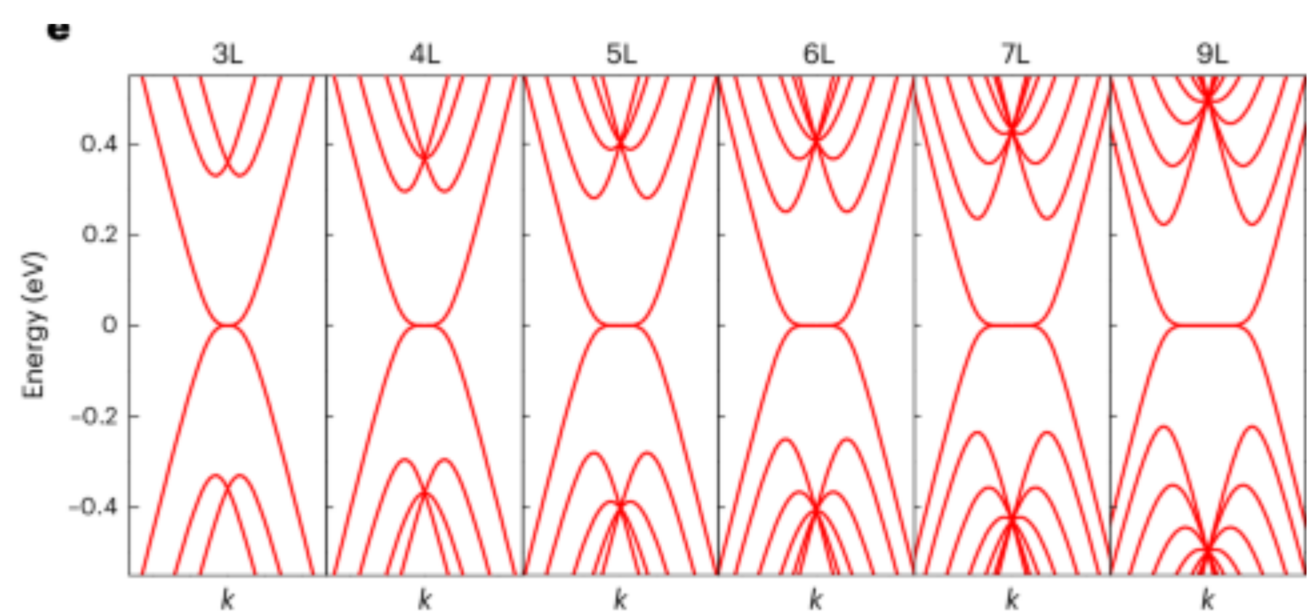




Dimensionality Matters: Changes in Electronic Structure




(a) simple hexagonal (AAA)
(b) Bernal (ABA)
(c) rhombohedral (ABC)
A
B
C
γ_0
γ_1
γ_2
γ_3
(d)
(e)
(f)
E-E$_F$ (eV)
1
0
-1
1.6 1.7 1.8
k$_{//}$ (Å^{-1})


e
3L
4L
5L
6L
7L
9L
Energy (eV)
0.4
0.2
0
−0.2
−0.4
k

Computational Challenges in 2D Modeling

Some Precautions

$$E[\rho] = T_s[\rho] + \int v_{ext}\rho(\mathbf{r})d\mathbf{r} + E_H[\rho] + E_{xc}[\rho]$$

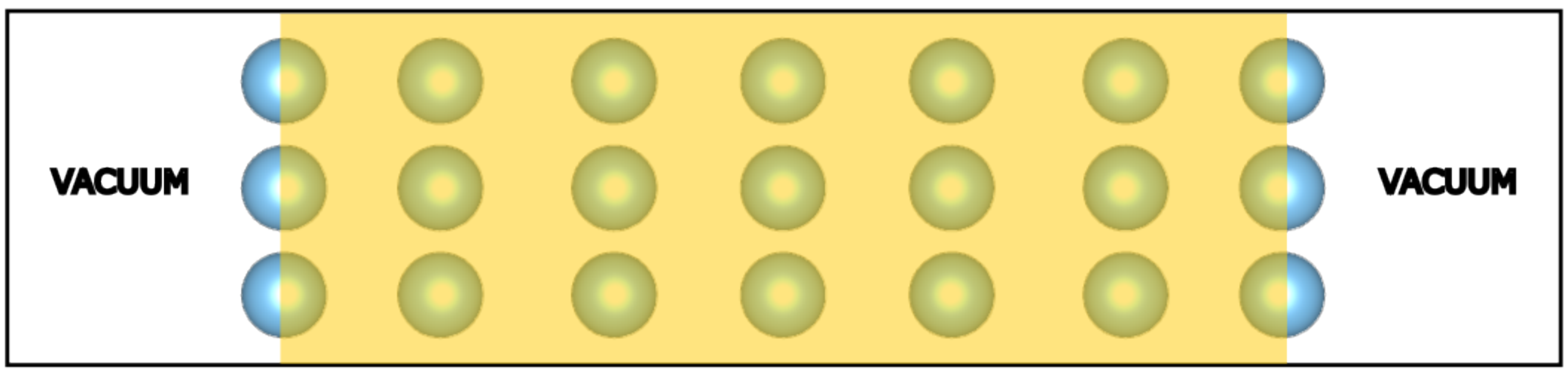
$$E_H[\rho] = \frac{1}{2}\int\int \frac{\rho(\mathbf{r})\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}d\mathbf{r}'$$

$$V_H[\rho] = \frac{\delta E[\rho]}{\delta\rho} = \int \frac{\rho(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}' \qquad \nabla^2 V_H(\mathbf{r}) = -4\pi\rho(\mathbf{r})$$

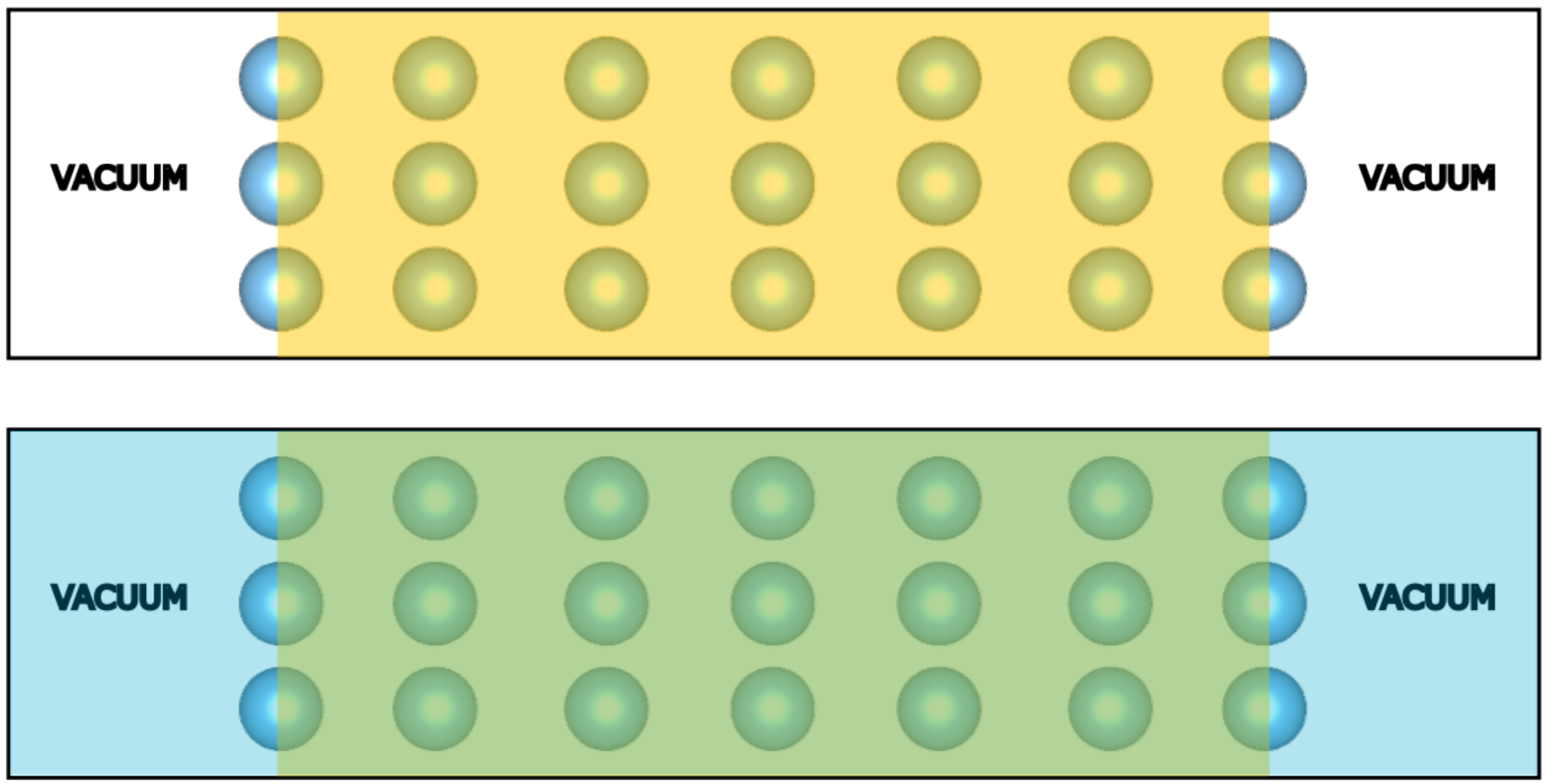
$$\rho(\mathbf{r}) = \sum_{\mathbf{G}} \rho(\mathbf{G})e^{i\mathbf{G}\cdot\mathbf{r}} \qquad V_H(\mathbf{r}) = \sum_{\mathbf{G}} e^{-i\mathbf{G}\cdot\mathbf{r}} \qquad \nabla^2 e^{-i\mathbf{G}\cdot\mathbf{r}} = -G^2 e^{-i\mathbf{G}\cdot\mathbf{r}}$$

$$V_H(\mathbf{G}) = \frac{4\pi\rho(\mathbf{G})}{G^2} \qquad V_H(\mathbf{G}=\mathbf{0}) = \frac{4\pi\rho(\mathbf{G})}{G^2} \qquad \rho(\mathbf{G}=0) = \int_\Omega \rho(\mathbf{r})d\mathbf{r} = q$$

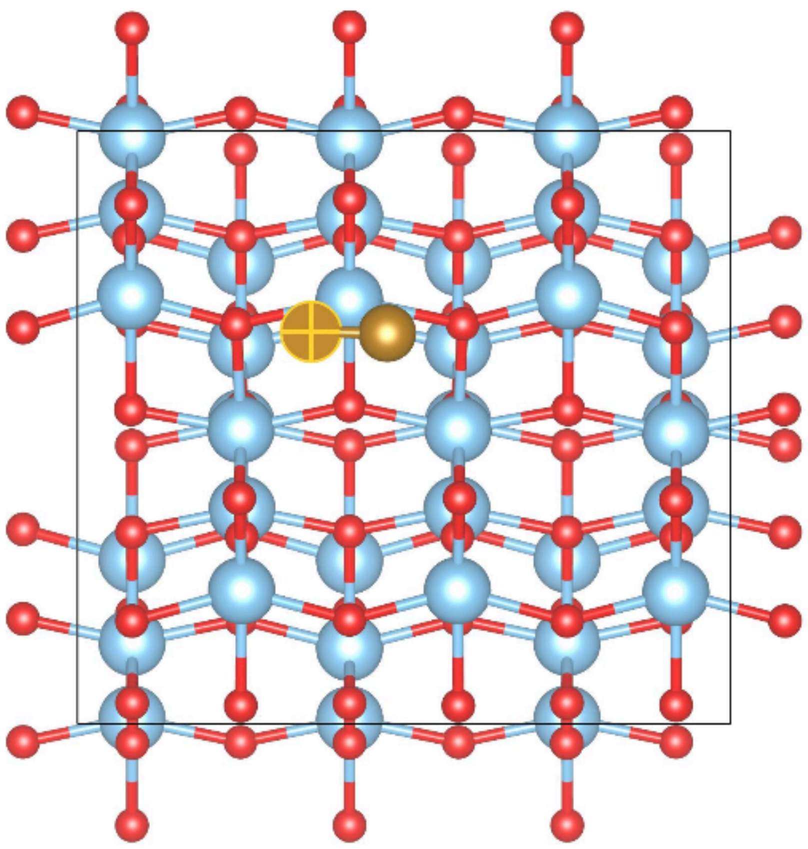
Some Precautions

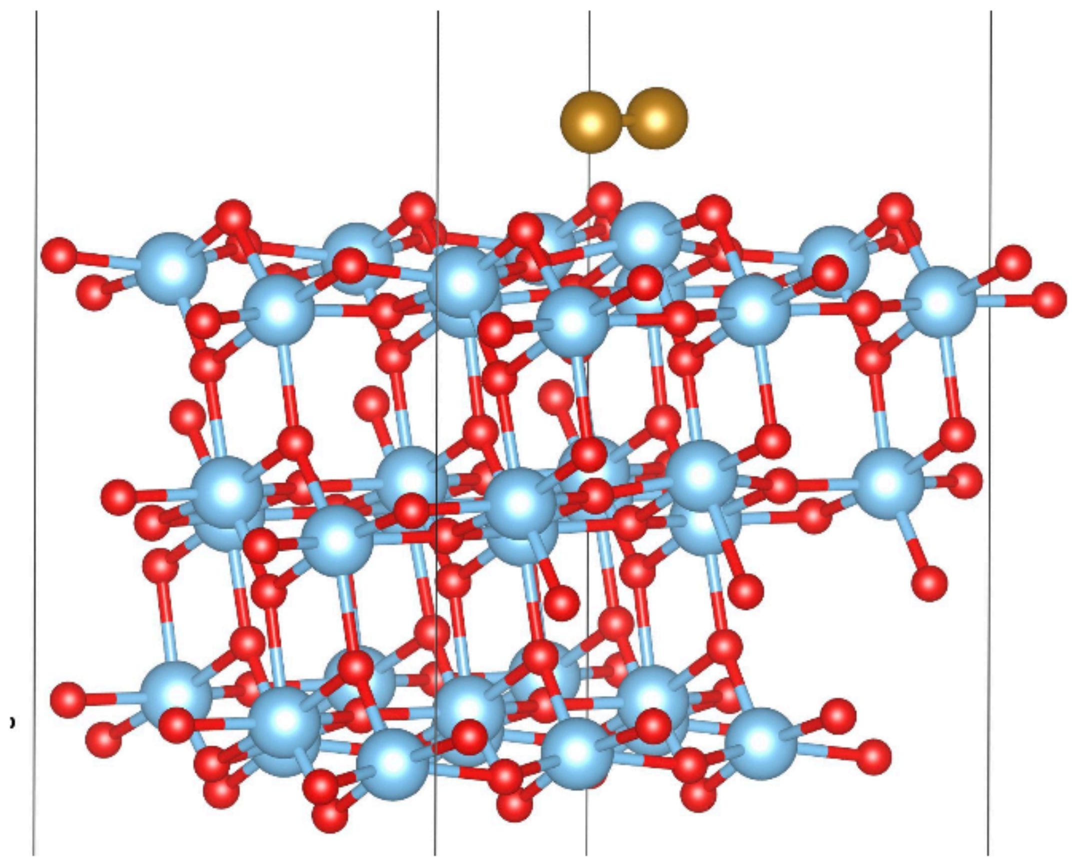


Some Precautions



Some Precautions





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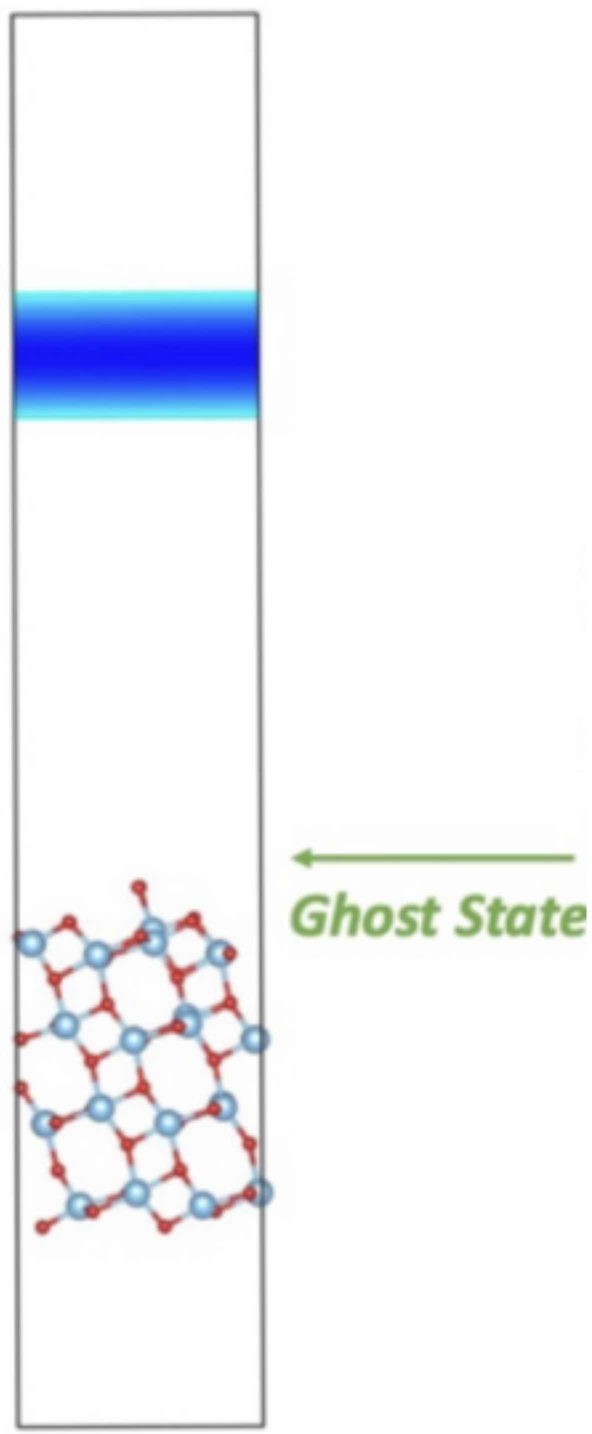

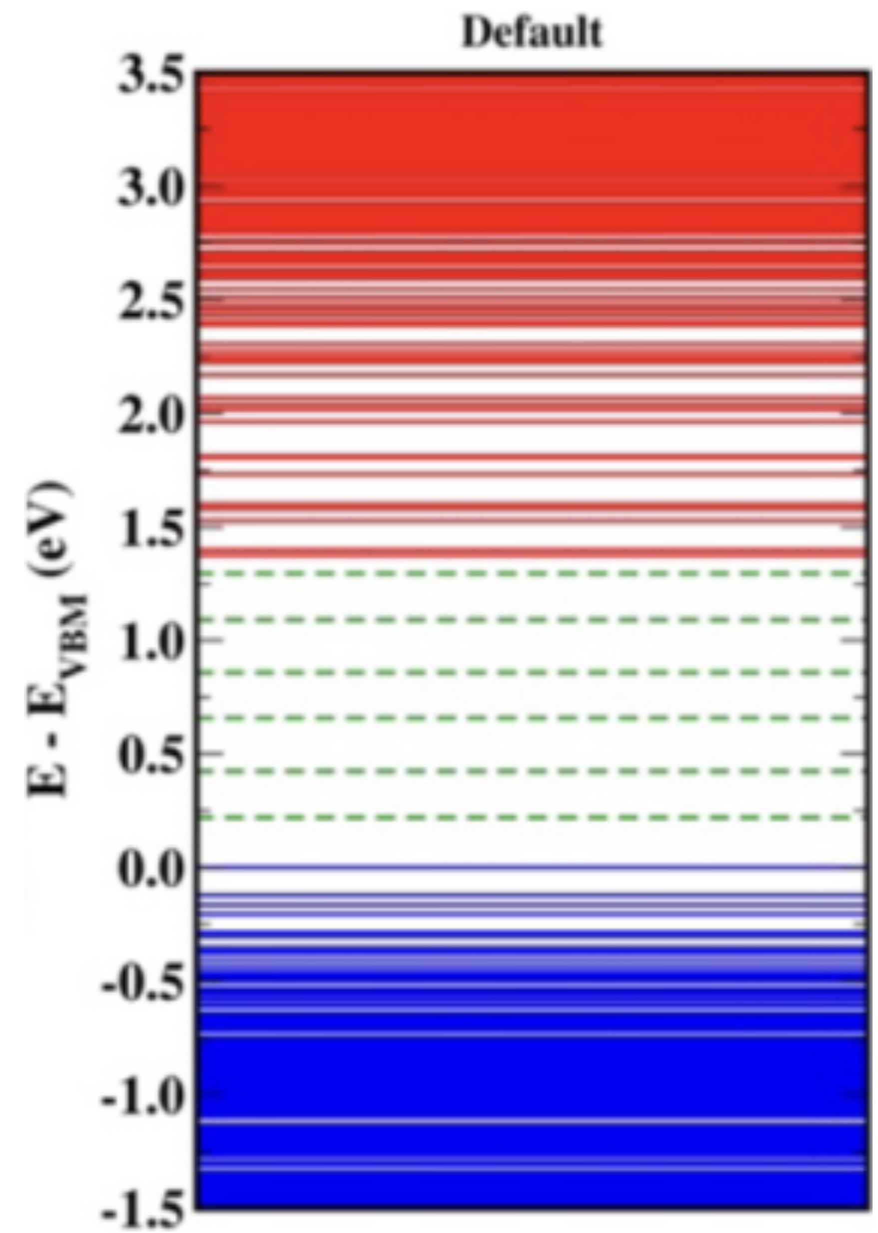

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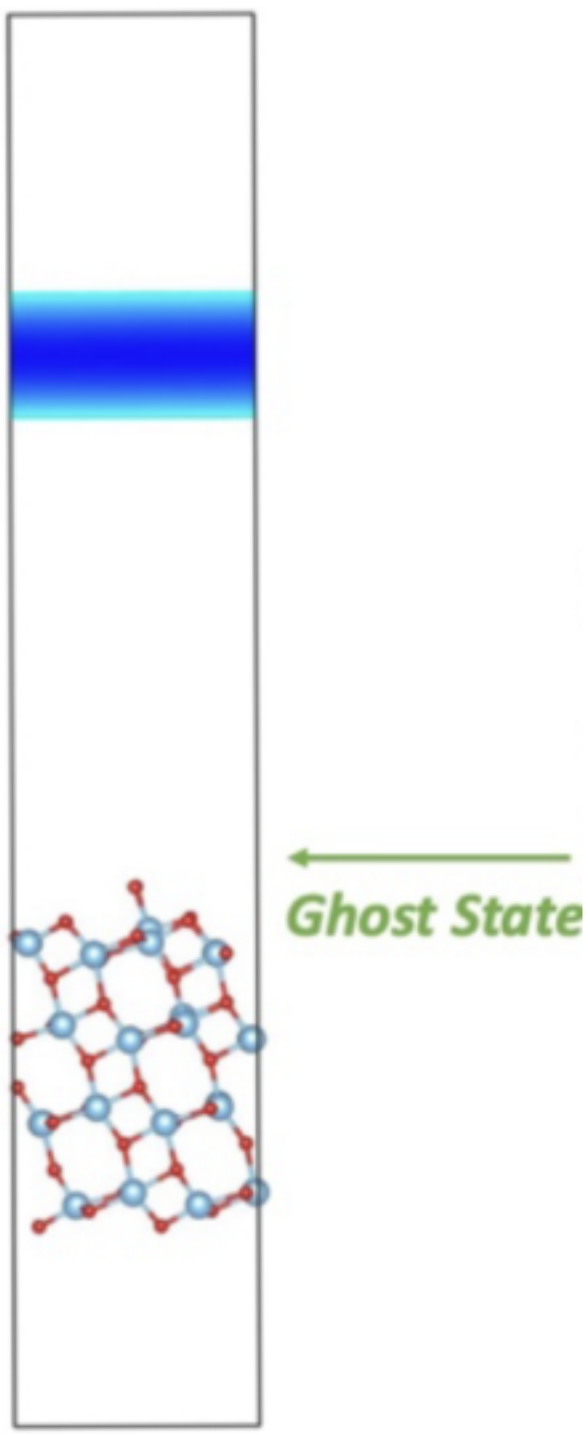
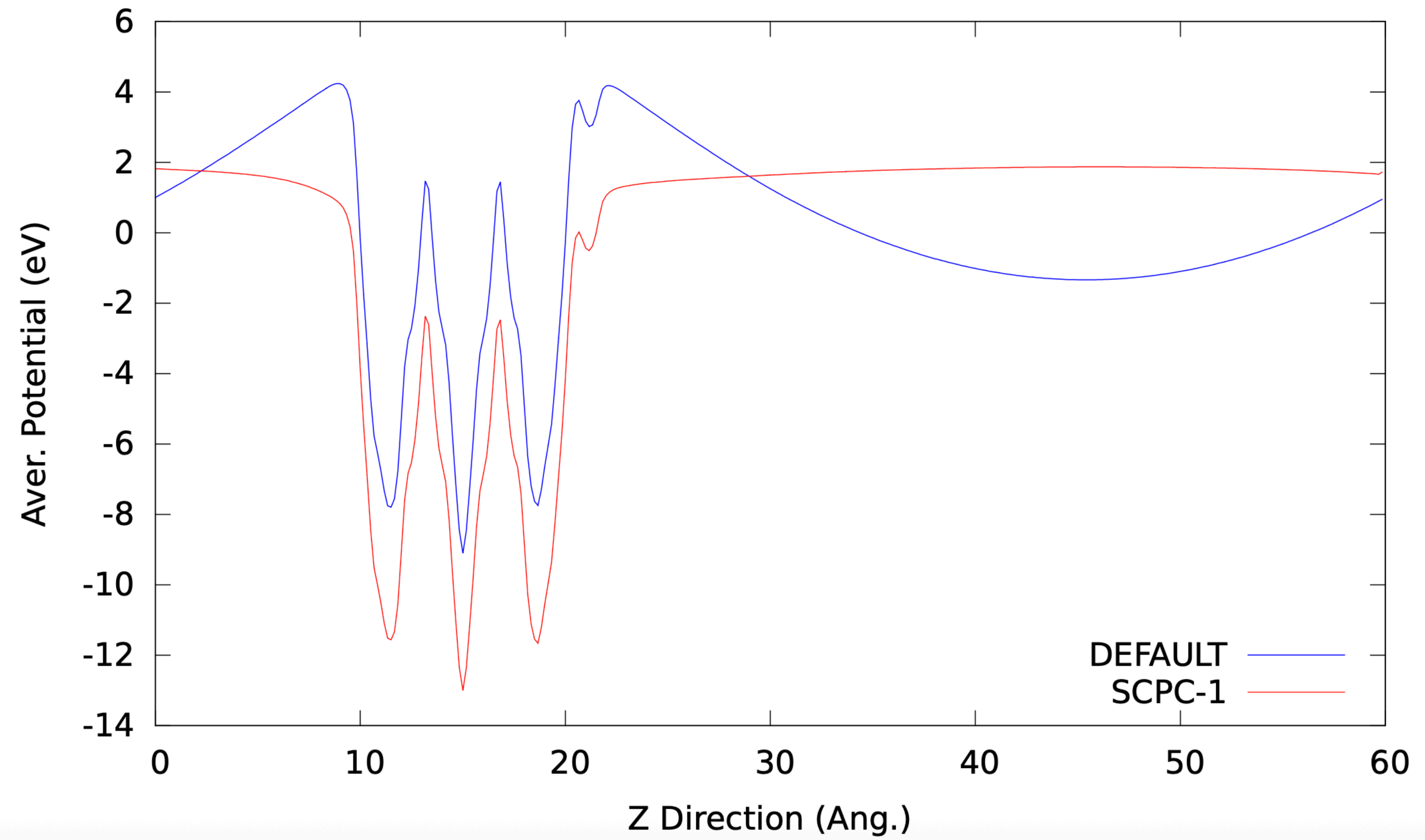
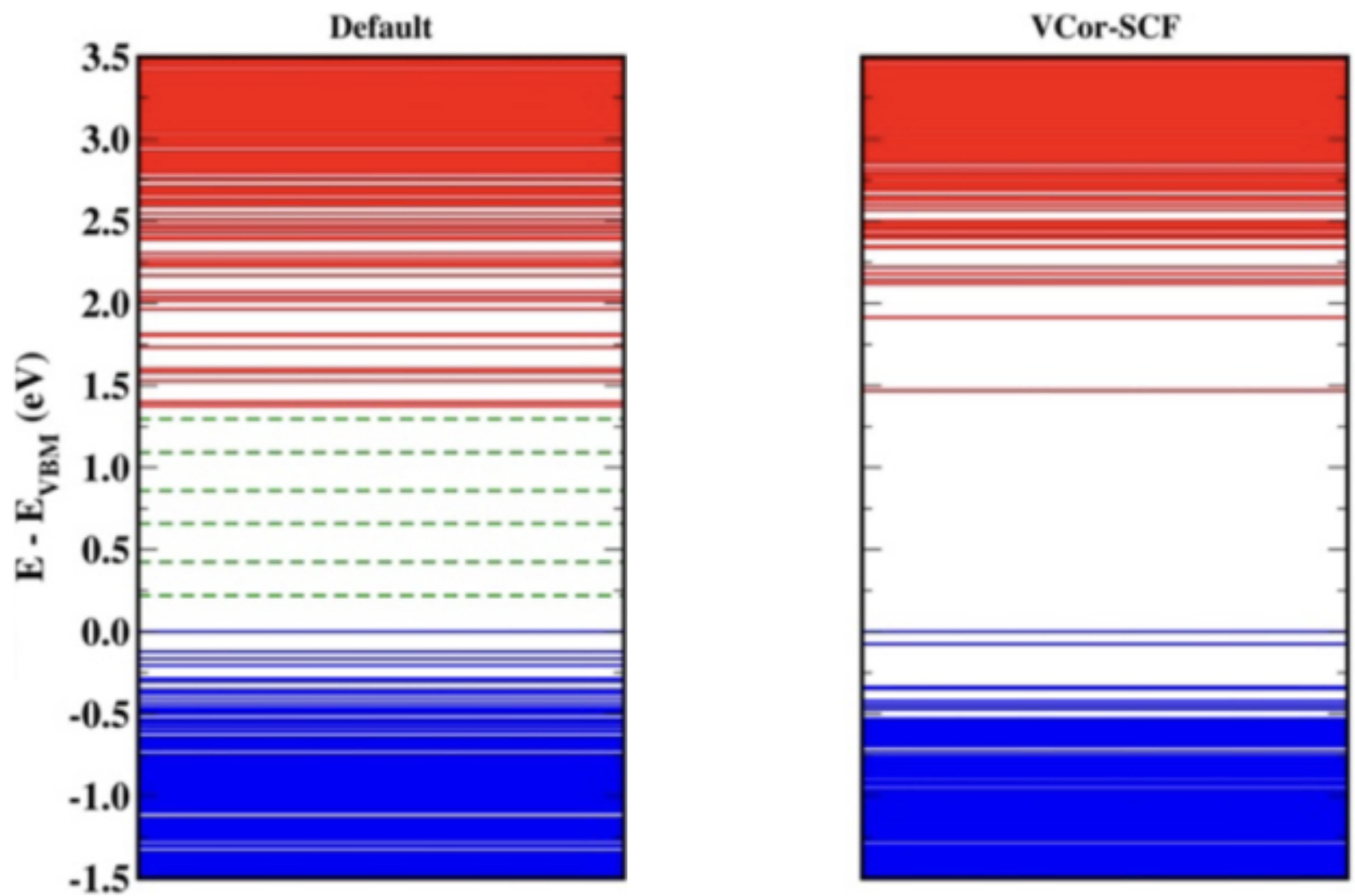




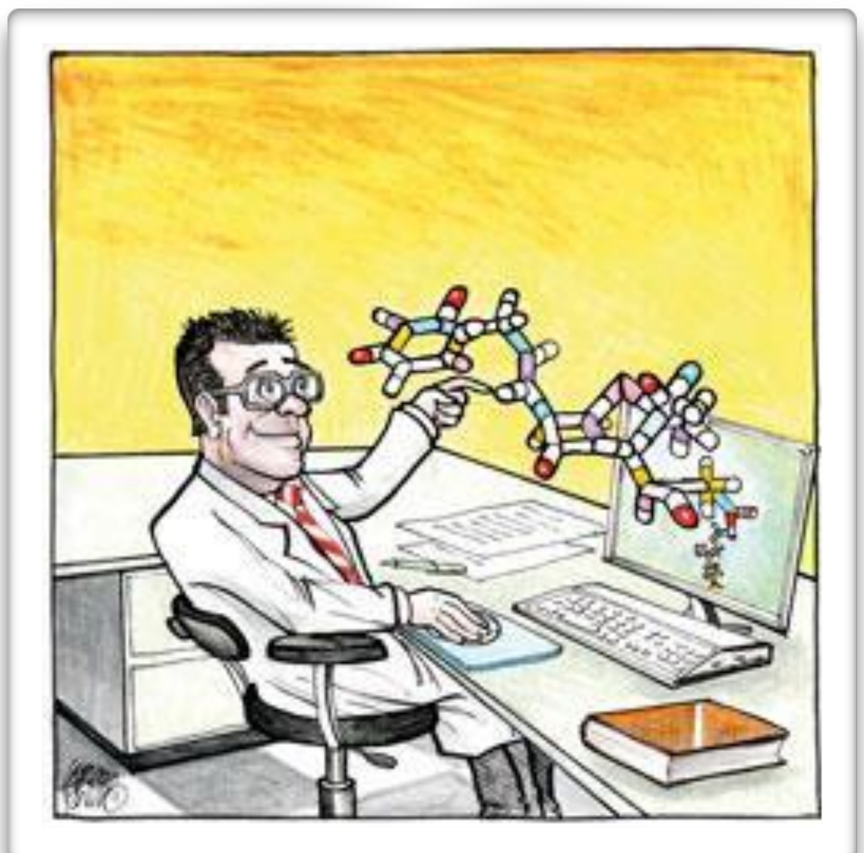
Some Precautions

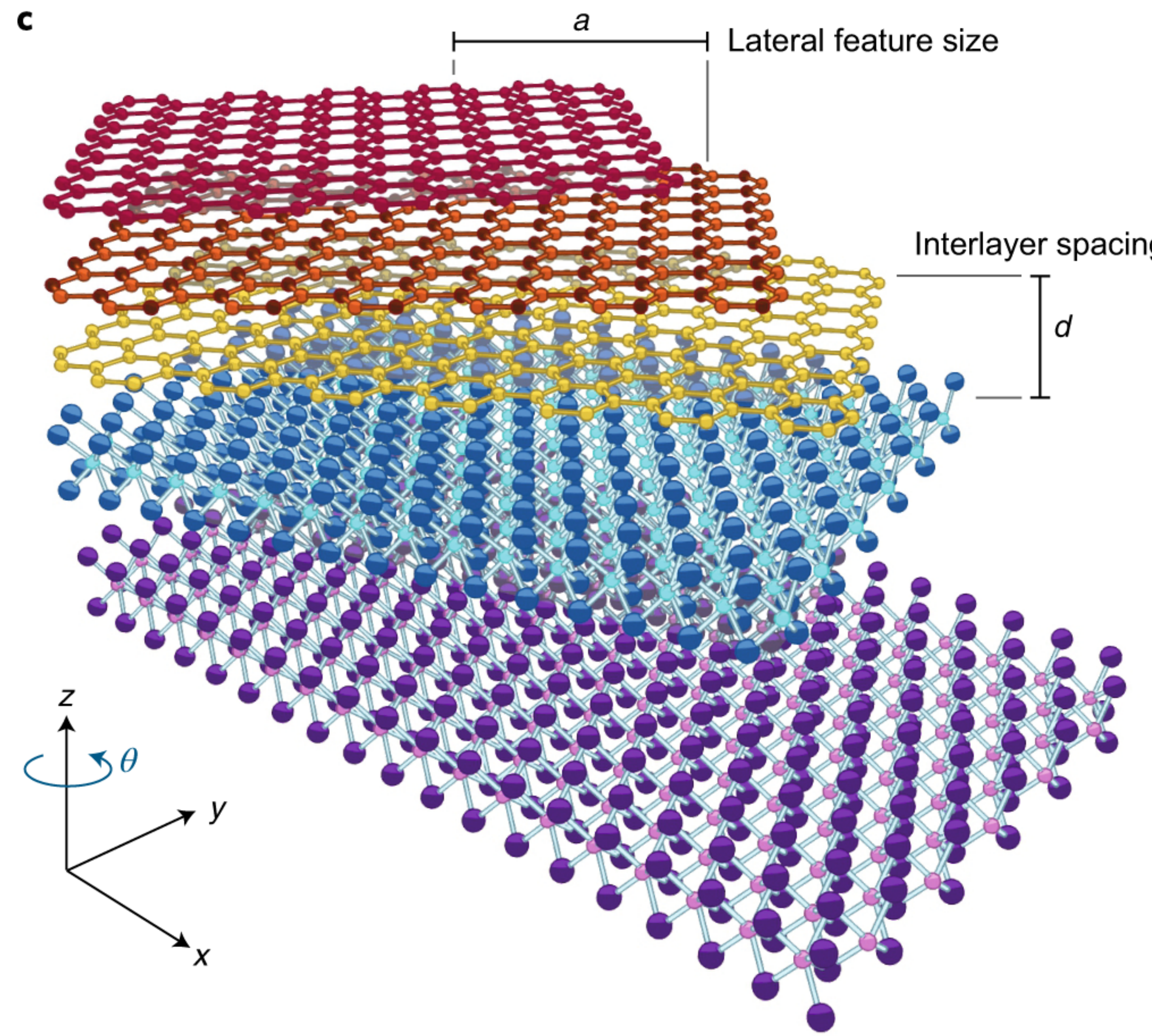


Hands On: From MgB2 to Graphite

- Step 0 - Structure
- **Step 1 - Structural Relaxation:** Optimizing lattice vectors and atomic positions for the monolayer.
- **Step 2 - Band Structure Calculation:** Computing and plotting the electronic bands to visualize the Dirac cone.

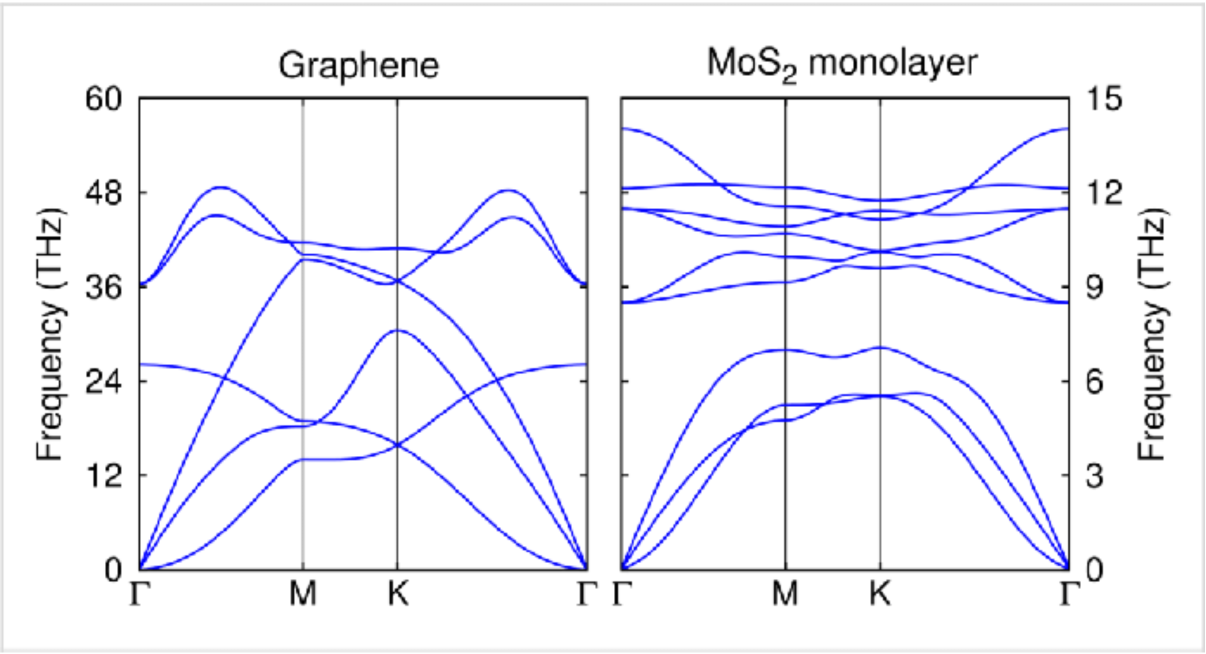
Is Imagination the Limit?

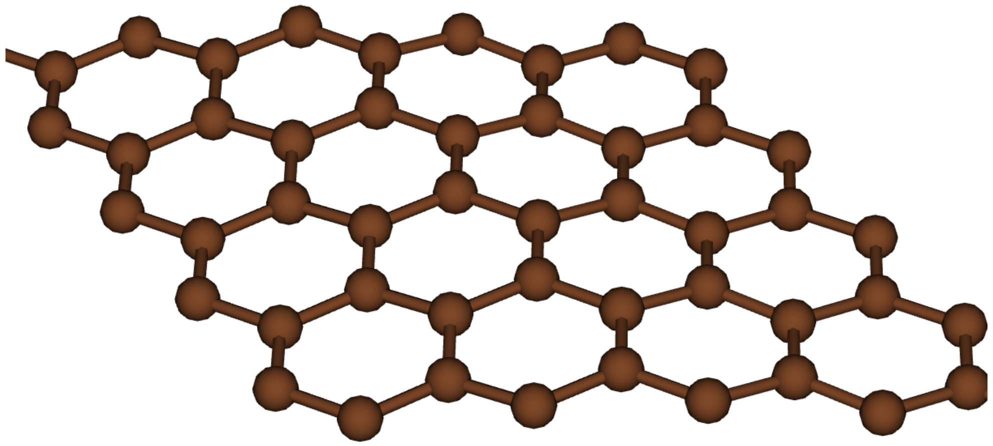




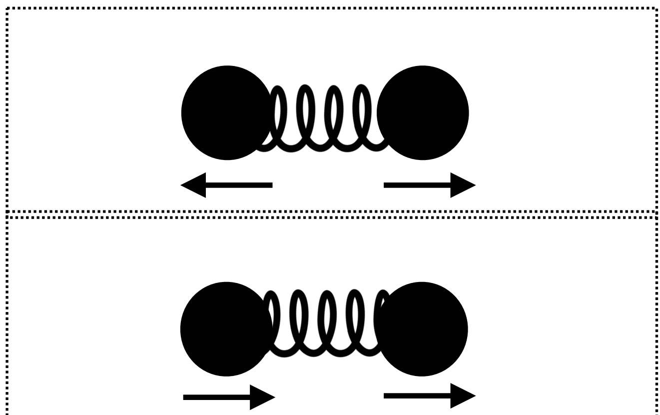
Now that we can isolate monolayers, can we theoretically exfoliate *any* bulk material or build whatever 2D lattice we want?

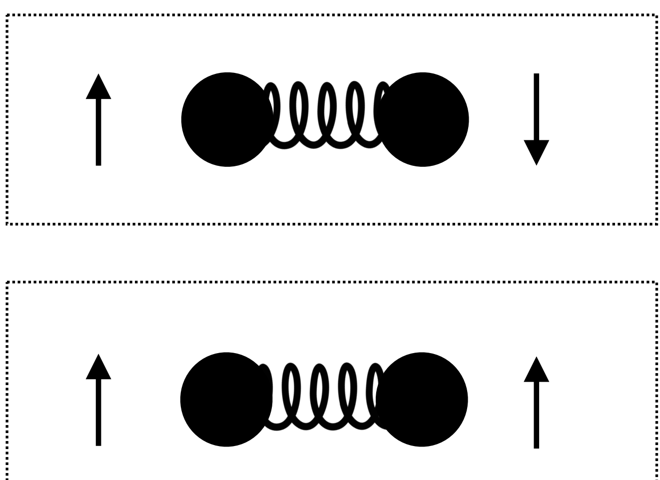
Phonons: The Ultimate Limit

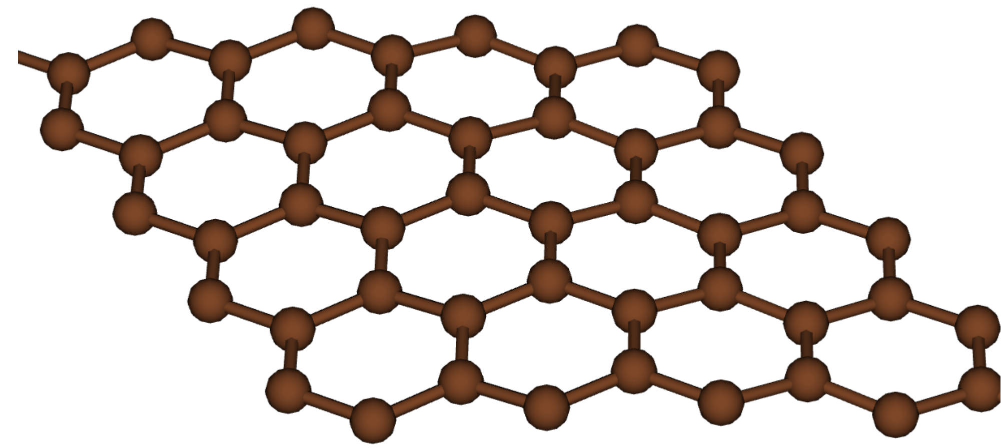


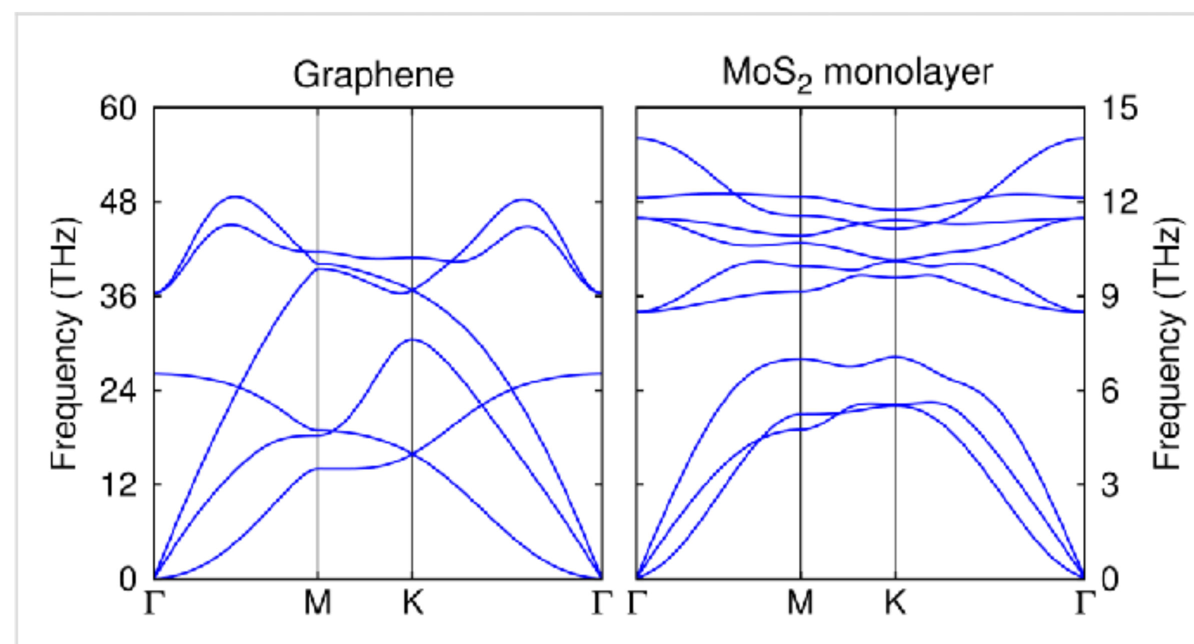


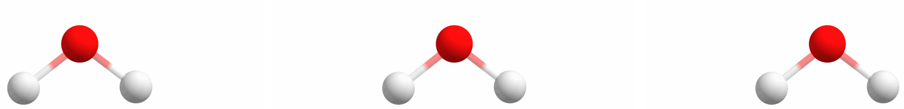
Phonons: Normal Modes of Vibration



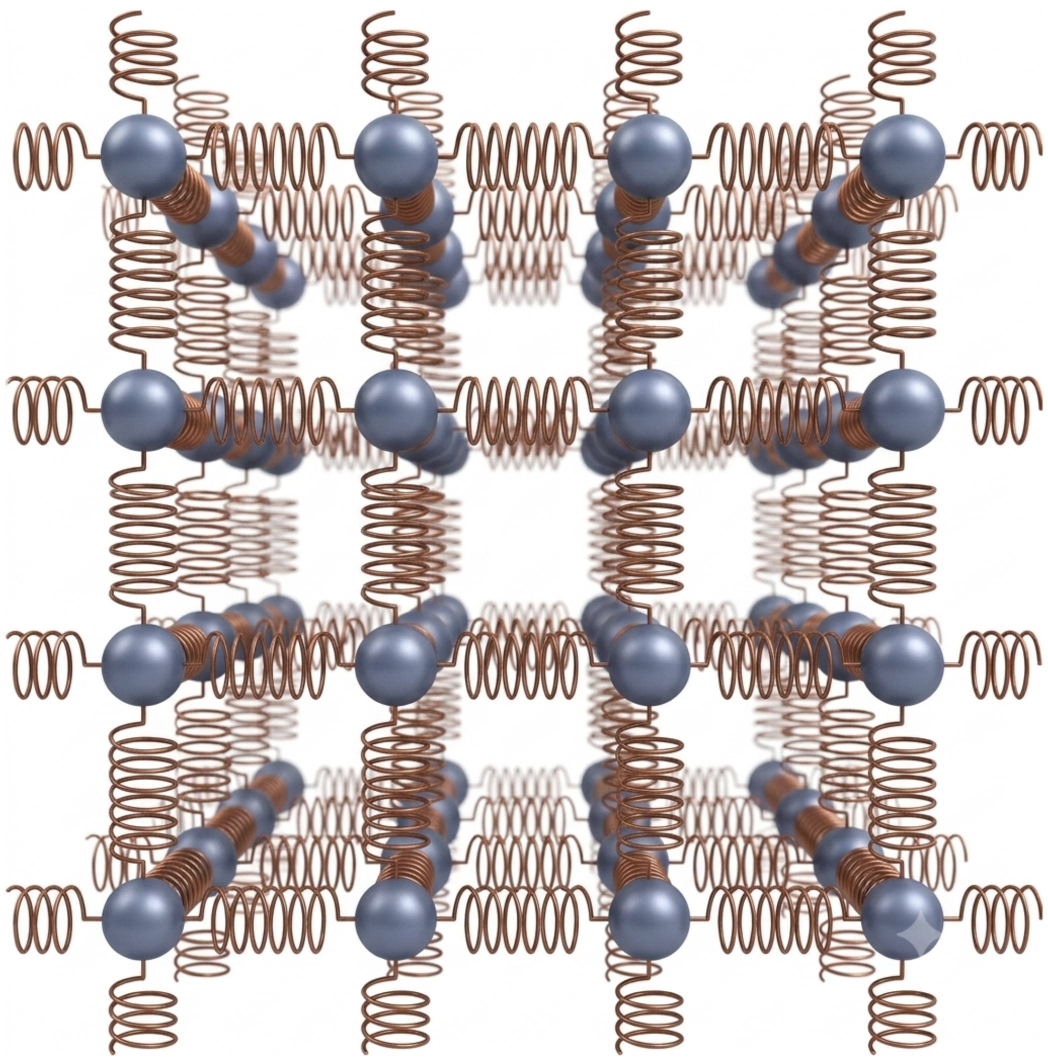


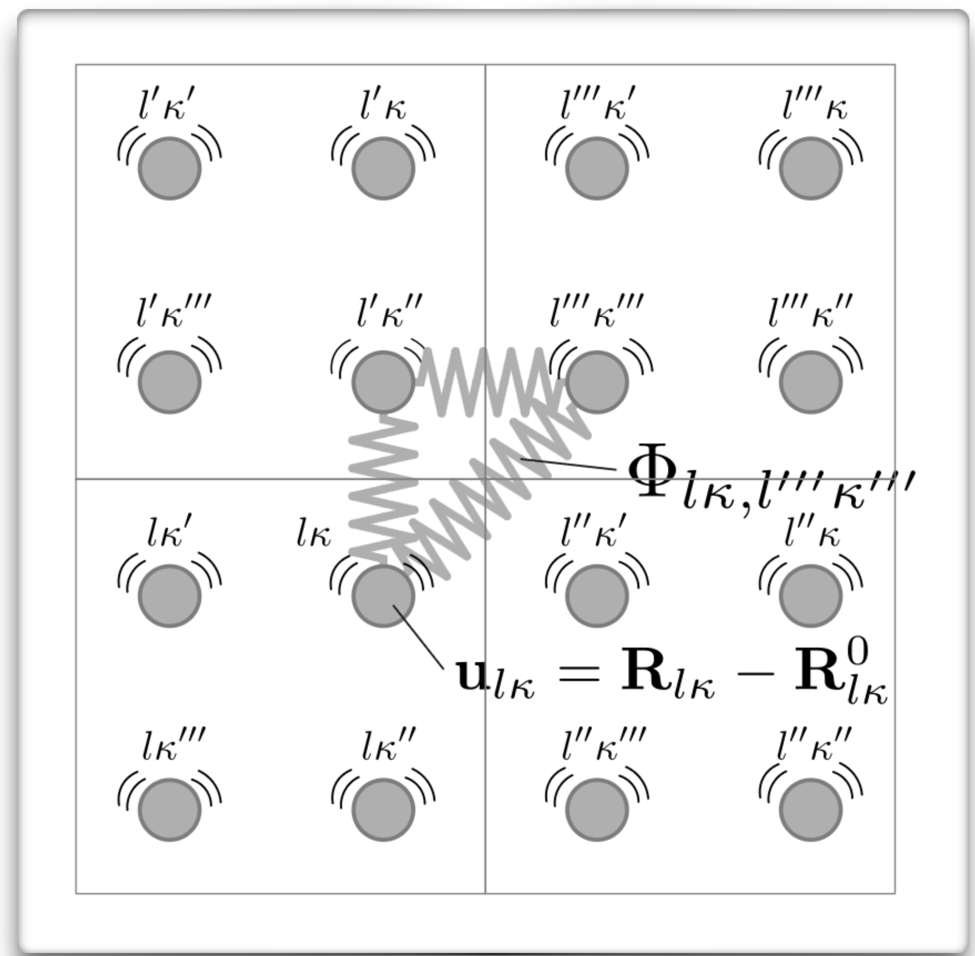




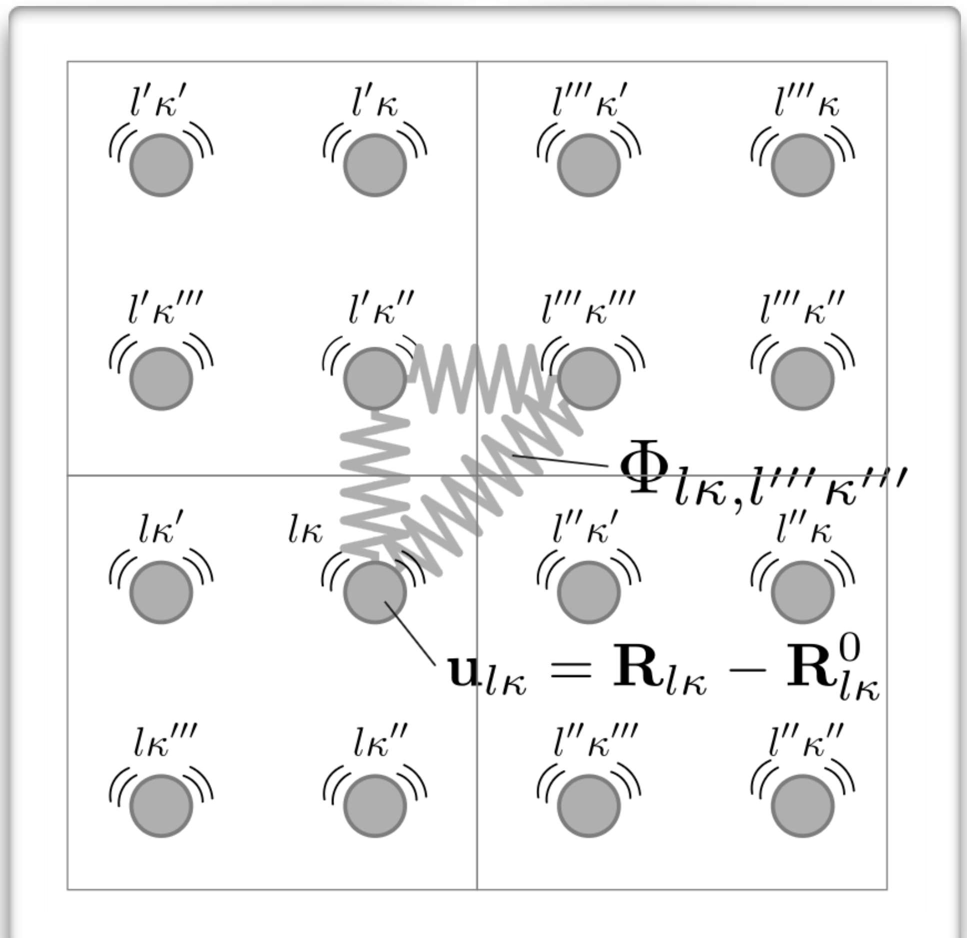


Phonon Calculation





Phonon Calculation



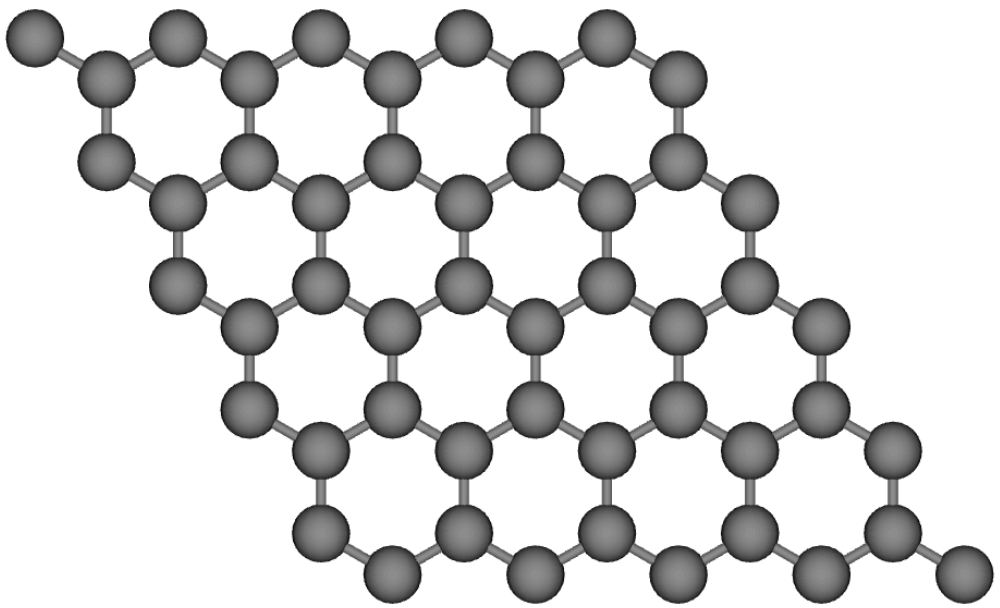
Supercell

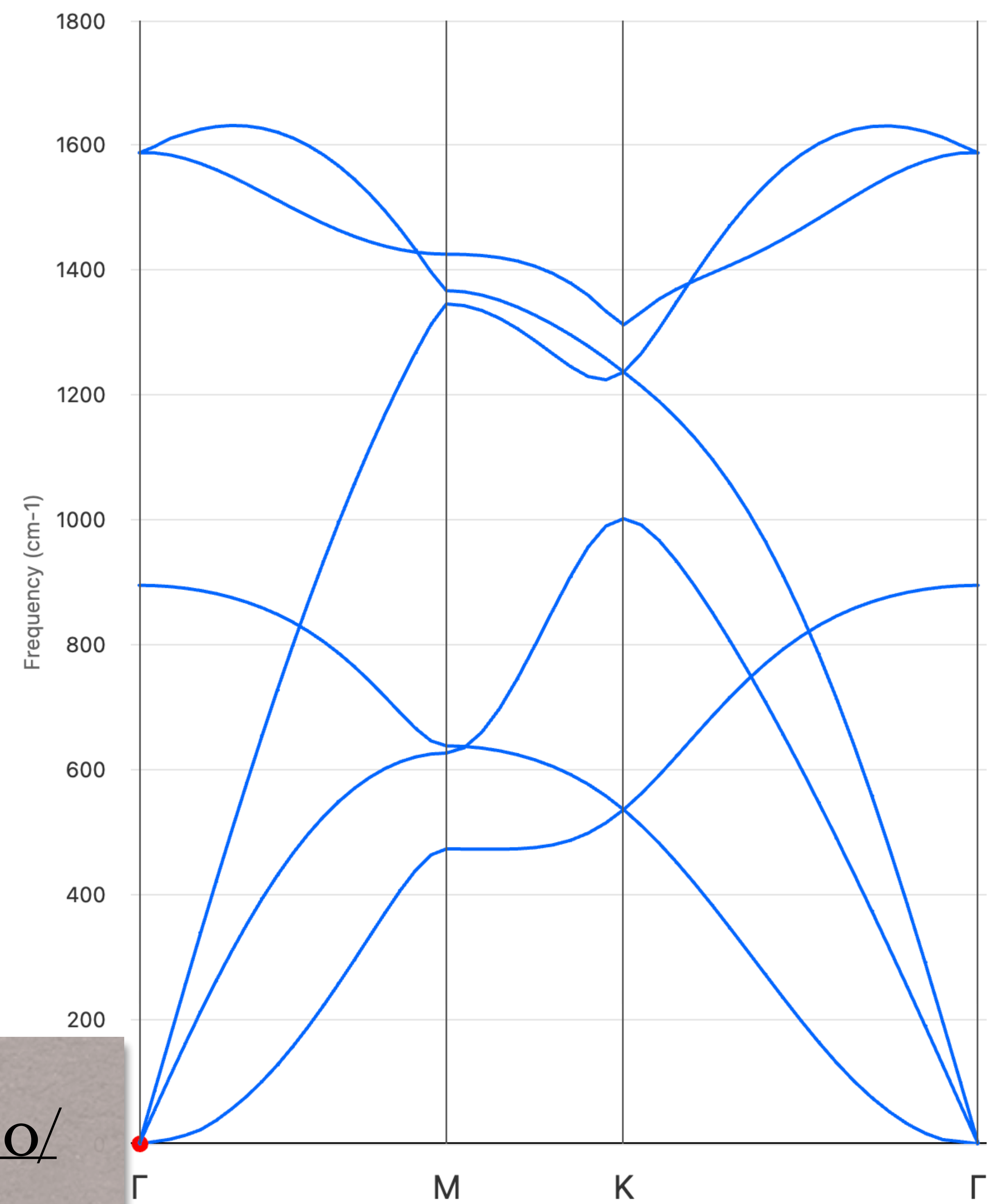
Calculation of **forces** on atoms in supercells with introduced **finite atomic displacements**

DFPT

Direct calculations of **force constants** at **arbitrary** wave vectors of primitive cell

Understanding Phonon Dispersions





https://henriquemiranda.github.io/phononwebsite/phonon.html